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Mediational analysis of severe retinal injury causation in children with acute closed head injury

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ABSTRACT

Background: While generally regarded as discriminating findings, elevating the probability of abusive head trauma, the association of severe retinal injuries with inertial biomechanics, shaking, and abuse has been challenged.
Objective: Evaluate five causational models of severe retinal injury utilizing multivariable mediational analysis.
Participants: Children, under age three-years, admitted to 18 pediatric intensive care units with acute, symptomatic, closed head injury who had not been injured in a car-crash and who had received an ophthalmologist's retinal examination.
Methods: Multivariable logistic regression, and multiple mediational analysis were applied to five causational models of severe retinal injury. Primary causes evaluated were inertial injury biomechanics, severe parenchymal brain injury, and provision of cardio-respiratory resuscitation. Results were reported as direct and mediated log-odds (β), with standard errors and p-values.
Results: The most explanatory model, having the greatest total effect ($\beta = 3.768$, $p \leq 0.001$) and direct effect ($\beta = 2.431$, $p \leq 0.001$), identified isolated inertial injuries as the primary cause of severe retinal injury. Additional indirect effects were mediated by radiographically evident diffuse parenchymal brain injury ($\beta = 0.834$, $p \leq 0.001$) and the presence of extra-axial blood on imaging ($\beta = 0.589$, $p \leq 0.001$).
Conclusions: Causational modeling of cross-sectional data in this population best supports inertial head injury biomechanics, above a threshold to cause diffuse parenchymal brain injury and extra-axial hemorrhage, as the most likely cause of severe retinal injury. Support for other causal pathways is statistically significant, but less robust.

1. Introduction

Retinal injury, hemorrhage and retinoschisis, is seen as a cardinal indicator of abusive head trauma, with substantial evidence base for its diagnostic significance (Bhardwaj et al., 2010; Binenbaum et al., 2009; Hymel et al., 2019; Hymel et al., 2022; Hymel et al., 2024; Maguire et al., 2009; Maguire et al., 2011; Maguire et al., 2013; Piteau et al., 2012). The genesis of retinal injury is generally believed to be vitreo-retinal traction during trauma initiated rotational movement of the child's head (Christian et al., 2018; Gjerde and Mantagos, 2021; Levin, 2010; Nadarasa et al., 2014). Some have argued that other causes of intracranial pathology are the predicate

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for retinal hemorrhage (Brook, 2024; Cohen and Scheimberg, 2009; Gabaeff, 2011; Geddes et al., 2003; Mack et al., 2009; Scheimberg et al., 2013; Squier et al., 2012; Thiblin et al., 2022). Mechanism is important, as it gives a clue to the plausibility of reported traumatic explanations, the possibility of non-traumatic causes and the nature of hypothesized abusive acts. Two recent publications utilized data from the Pediatric Brain Injury Research Network studies to analyze associations with retinal hemorrhage (Brook, 2024; Sokoloff et al., 2024). Sokoloff et al. argued that accelerative trauma was the cause of ophthalmologist identified severe retinal injury (Sokoloff et al., 2024). Brook argued that the occurrence of severe retinal injury reflects the general severity of neurotrauma, as demonstrated by radiographic diffuse hypoxic ischemic encephalopathy or brain swelling (HIE), and that abusive head trauma, treated as shaking, was irrelevant (Brook, 2024). Both studies advanced their hypotheses using univariable tests of association.

Given the strong co-variance between findings in traumatic brain injury (Boos et al., 2022; Burkhart et al., 2015; Morad et al.,

Table 1

Variables used in analyses (short name used in text for conciseness) and definition of the variable.

Variable	Definition
Abusive Head Trauma (abuse)	Consistent with the PediBIRN series of studies, *a priori* criteria for abusive head trauma are met when any one of the following is true (Hymel et al., 2024) - The primary caregiver admitted abusive acts - Abusive acts by the primary caregiver were witnessed by an unbiased, independent observer - The primary caregiver specifically denied that the preambulatory child in his/her care had experienced any head trauma - The primary caregiver provided an account of the child's head injury event that was clearly historically inconsistent with repetition over time - Further workup confirmed the presence of two or more of the following injuries, considered moderately or highly suspicious for abuse • Classic metaphyseal lesion fracture(s) or epiphyseal separation(s) • Rib fractures • Fractures of the scapula or sternum • Fractures of digits • Vertebral body fracture(s) dislocation(s) or fractures of spinous process(es) • Scalding burn(s) with uniform depth, clear lines of demarcation, and paucity of splash marks • Confirmed intra-abdominal injuries
Child age in months (age)	Child's age in months at presentation
Respiratory compromise prior to ED arrival (respiratory compromise)	Provision of respiratory support prior to arrival at ED
Circulatory compromise prior to ED arrival (circulatory compromise)	Provision of circulatory support prior to arrival at ED
Clinical indications of a hypoxic or ischemic challenge (clinical hypoxia or ischemia)	Either respiratory compromise or circulatory compromise reported
Seizure	Seizures reported prior to arrival at ED
Altered consciousness prior to ED arrival (altered consciousness)	Loss or alteration of consciousness prior to arrival at ED
Duration of altered consciousness and occurrence of deterioration in mental state (duration of altered consciousness)	Duration of altered consciousness was used as one measure of parenchymal brain injury (Genarelli et al., 1982; Ommaya and Gennarelli, 1974). 0 = No reported altered consciousness 1 = altered consciousness resolved prior to ED 2 = altered consciousness lasting <24 h 3 = altered consciousness lasting >24 h 4 = altered consciousness lasting >24 h with deterioration
Subarachnoid hemorrhage	Any subarachnoid hemorrhage(s)
Contact associated subdural hemorrhage (contact subdural)	Subdural hemorrhage(s) or collection(s) were presumed to be due to local contact forces when they were unilateral and not interhemispheric
Inertia associated subdural hemorrhage (inertial subdural)	Subdural hemorrhage(s) or collection(s) were presumed to be due to inertia when they were bilateral or interhemispheric (Genarelli et al., 1982; Genarelli and Thibault, 1982; Ommaya and Gennarelli, 1974)
Extra-axial hemorrhage (extra-axial blood)	Subarachnoid hemorrhage or subdural hemorrhage in any distribution
Intra-parenchymal injury, cortical (cortical brain injury)	Brain contusion(s), laceration(s), or hemorrhage(s) that are limited to the cortical brain
Intra-parenchymal injury, subcortical (subcortical brain injury)	Brain contusion(s), laceration(s), or hemorrhage(s) that involve the subcortical (or deeper) brain
Focal hypoxic-ischemic injury or edema (focal HIE)	Brain hypoxia, ischemia, or swelling that is unilateral and limited to the cortical brain
Diffuse hypoxic-ischemic injury (diffuse HIE)	Brain hypoxia, ischemia, or swelling that is bilateral OR involves the subcortical (or deeper) brain
Inertial injury biomechanics (inertial injury)	Experimental literature has linked traumatic alterations of consciousness and subdural hemorrhage due to bridging vein injury to inertial loading of the head (Genarelli et al., 1982; Genarelli and Thibault, 1982; Ommaya and Gennarelli, 1974). Inertial loading was inferred in the presence of altered consciousness, inertial subdural hemorrhage or both
Contact injury biomechanics (contact injury)	Any craniofacial soft tissue injury, skull fracture and/or epidural hematoma reported
Isolated inertial injury biomechanics (isolated inertial injury)	Inertial injury in the absence of contact injury

2002), we believed multivariable analysis was necessary to distinguish between competing causational theories of severe retinal injury. We hypothesized that multivariable statistical methods would support inertial injury as the primary antecedent of severe retinal injury, while also supporting injury severity as an associated finding. We examined associates of severe retinal injury by two methods: multivariable regression and multiple mediational analysis. While the Sokolof and Brook studies have reported statistical associations, the clinical and legal arguments that stem from these findings often invoke causational interpretations. Standard multivariable regression methods examine associations adjusted for covariates, but do not directly test causational mechanisms. In contrast, multiple mediational analysis is specifically designed to evaluate potential causational pathways. This method quantifies both the direct effect of a proposed cause on an outcome and the indirect effect mediated through intermediaries, thereby offering a more comprehensive understanding of how and why an effect occurs, supporting causational interpretation. Multiple mediational analysis is commonly performed in a longitudinal framework, where causes, mediators and outcomes are assessed sequentially over time. However, it is also applicable to cross-sectional data, provided the variables are assessed within a narrow time window and the hypothesized temporal ordering is theoretically justified (Hayes, 2017; Preacher, 2015).

2. Methods

2.1. Overview

This was a secondary analysis of a subpopulation of the combined Pediatric Brain Injury Research Network cross-sectional data sets. We used two methods, multivariable logistic regression and multiple mediational analysis, to analyze the causality of findings associated with severe retinal injury. The Institutional Review Boards at Case Western Reserve and Baystate Medical Center determined that this study was not human subject research. Performance conformed to the requirements of the US Health Insurance Portability and Accountability Act of 1996,

2.2. Subjects

Data came from fully de-identified data sets acquired in the Pediatric Brain Injury Research Network derivation, validation and implementation studies. Detailed descriptions of the methods used in the studies are available elsewhere (Hymel et al., 2013; Hymel et al., 2021; Hymel et al., 2024). Patient records contain 38 uniform, prospective, clinical, historical, and radiological data fields regarding 973 acutely symptomatic patients under three years of age with CT- or MRI-confirmed closed head injuries hospitalized for intensive care at one of 18 academic medical centers between February 2011 and March 2021. Victims of motor vehicle collisions and patients with pre-existing brain abnormalities were excluded. The subset of 701 records with documented results of an ophthalmologist's examination formed the sample for the current study.

2.3. Predictor variables

Predictor variables were either reported by Pediatric Brain Injury Research Network sites or derived from directly reported variables (Table 1). Derived variables included combinations, structured ordinals, and proxies for injury concepts as elaborated in the discussion. The prospective study design in all three studies facilitated the capture of uniform, complete data regarding >90 % of eligible subjects based on monthly audits.

2.4. Outcome variable

The outcome for all analyses was the presence or absence of severe retinal hemorrhage "described by an ophthalmologist as dense, extensive, covering a large surface area, and/or extending to the ora serrata" and/or "retinoschisis confirmed by an ophthalmologist," collectively referred to as "severe retinal injury."

2.5. Statistical analyses

Multivariable logistic regressions evaluated associations between severe retinal injury and the variables abuse, age, respiratory compromise, circulatory compromise, seizure, altered consciousness, subarachnoid hemorrhage, contact subdural, inertial subdural, cortical brain injury, subcortical brain injury, focal HIE, and diffuse HIE. We fitted a full model, including all variables, and a model with selected variables based on the backward selection strategy. Results were stated as odds ratios, 95 % confidence intervals (CI), and *p*-values.

We conducted multiple mediational analysis to explore hypotheses of primary causational and mediational pathways in the formation of severe retinal injury. Five analyses were performed with severe retinal injury as the outcome variable and five candidate primary causes: isolated inertial injury with no indicators of contact injury, inertial injury with or without indicators of contact injury, diffuse HIE compared to the absence of any HIE, diffuse HIE compared to absent or focal HIE, and circulatory or respiratory compromise requiring resuscitation prior to ED arrival as clinical indicia of hypoxia or ischemia. Multiple mediators (diffuse HIE, focal HIE, clinical hypoxia or ischemia, seizure, an ordinal ranking for duration of altered consciousness, and extra-axial blood in any distribution) were included in models when they were not the candidate primary cause (Table 2). In the multiple mediational analysis framework, logistic regression was used for the primary outcome analysis, and multivariable logistic regression was applied for multiple mediator regression. The direct, indirect, and total effects – reported as the parameter estimate in the scale of log-odds (β) along with their standard errors (SE) and p-values – as well as the sample sizes for subjects with the variable present and absent are presented in Fig. 1.

All analyses were performed using RStudio version 4.3.3, utilizing the "glm" function from the "stats" package for logistic regression and the "mma" function from the "mma" package for multiple mediational analysis. Notably, the selection of mediators for the final mediation analysis was based on two conditions: 1. the variable is significantly correlated with the primary cause; 2. the variable is associated with the outcome, given that all other relevant factors are included in the model. For both conditions, the significance level was set at 0.1 (the default option in the "mma" function). For all other hypothesis tests, a P-value <0.05 was used to indicate statistical significance.

3. Results

Multivariable logistic regression identified positive associations between severe retinal injury and the variables abuse, circulatory compromise, seizure, contact subdural, inertial subdural, and diffuse HIE. A negative association existed for age. In the select model all retained variables, other than respiratory compromise, were significantly associated with severe retinal injury (Table 3).

All multiple mediational analyses demonstrated direct and mediated effects (Fig. 1). The strongest effects were for isolated inertia, with total $\beta = 3.731$ and direct $\beta = 2.431$, 35 % mediated by diffuse HIE and extra-axial blood. A similar total effect was demonstrated for inertial injury with or without contact injury (total $\beta = 3.297$, direct $\beta = 1.701$), but 48 % mediated by extra-axial blood, clinical hypoxia or ischemia, diffuse HIE, and seizure, in decreasing order. Diffuse HIE had less effect (total $\beta = 2.215$, direct $\beta = 0.936$ compared to no HIE; total $\beta = 2.104$, direct $\beta = 0.889$ compared to absent or focal HIE). More than 50 % of the total effect was mediated by duration of altered consciousness, clinical hypoxia or ischemia, extra-axial blood, and seizure. Clinical hypoxia or ischemia had the least effect (total $\beta = 1.992$, direct $\beta = 0.604$), nearly 70 % mediated by duration of altered consciousness, diffuse HIE, seizure, and extra-axial blood.

4. Discussion

Traumatic injuries provide evidence of their biomechanical and pathophysiological antecedents. Physicians must consider inferred antecedents, alongside provided or conjectured history and differential diagnoses to evaluate concerns for child physical abuse. In that light, identifying the antecedent cause of severe retinal injuries assumes substantial importance; particularly for two commonly accepted postulates: severe retinal injury has diagnostic value for child abuse, and manual shaking of an infant is an important

Table 2
Plan for multiple mediational analyses with the candidate primary cause (column 1), a description of subjects to which subjects with the candidate primary cause were compared (column 2) and a list of studied mediators through which the primary cause may have acted (column 3).

Primary exposure	Comparator	Potential Mediator
Isolated inertial injury	Subjects with contact injury and without inertial injury	Diffuse HIE Focal HIE Clinical hypoxia or ischemia Duration of altered consciousness Seizure Extra-axial blood
Inertial injury with or without contact injury	Subjects with contact injury and without inertial injury	Diffuse HIE Focal HIE Clinical hypoxia or ischemia Duration of altered consciousness Seizure Extra-axial blood
Diffuse HIE	Subjects with no HIE in any location	Clinical hypoxia or ischemia Duration of altered consciousness Seizure Extra-axial blood
Diffuse HIE	Subjects without diffuse HIE (includes those with focal HIE or no HIE in any location)	Clinical hypoxia or ischemia Duration of altered consciousness Seizure Extra-axial blood
Clinical hypoxia or ischemia	Subjects without clinical hypoxia or ischemia (neither respiratory compromise nor circulatory compromise)	Diffuse HIE Focal HIE Duration of altered consciousness Seizure Extra-axial blood

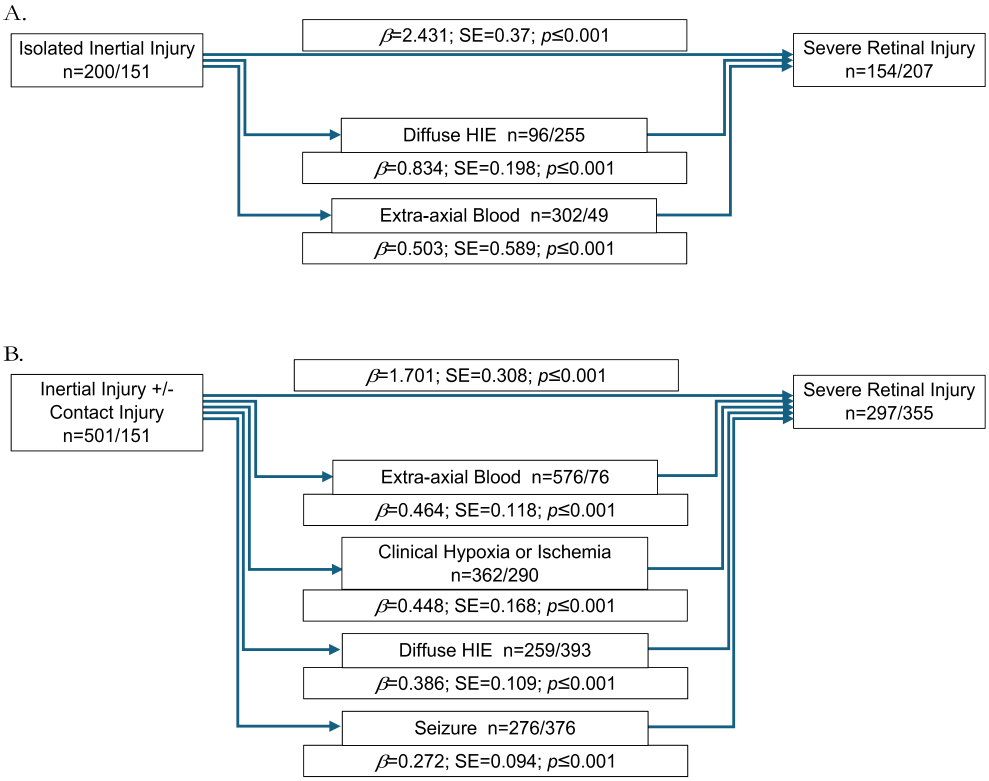

Fig. 1. Multiple mediational analyses, results of structural equation models.
For each panel, A through E, the candidate primary cause is to the left, outcome to the right and candidate mediators below and in the center. The n of subjects for which the variable is positive/negative occur alongside each variable. The log-odds (β), standard error (SE), and p-value for the associations occur centrally with indications of direct causation at the top and indications of mediated causation below each listed mediator.

mechanism of abusive head trauma.

4.1. Implications of multivariable logistic regression

In our first analyses, the full multivariable regression demonstrated that abuse, age, circulatory compromise, seizure, altered consciousness, contact subdural, inertial subdural, and diffuse HIE were significant associates of severe retinal injury (Table 3. Column 3). In our selected model, age, circulatory compromise, seizure, altered consciousness, contact subdural, inertial subdural, and diffuse HIE remained significant associates (Table 3. Column 4). These findings suggest that tests of causation for severe retinal injury should retain these variables in analysis, rather than testing a single association in a univariable analysis.

4.2. Rationale for multiple mediational analysis and selection of variables

Multiple mediational analysis allowed us to use statistical methods designed to explore causational relationships, while considering that causation may not follow a simple one-to-one pathway. A proposed primary cause may exert a direct effect on severe retinal injury, or it may act indirectly through one or more mediators. A mediator may represent an intervening step in a pathophysiologic chain, or an indicator of specific circumstances under which the cause produces severe retinal injury. Because the multiple mediational analysis requires one primary cause to be designated, while allowing for the inclusion of multiple mediators, we evaluated five models, testing associates that have been proposed as causes of severe retinal injury in literature and/or in court.

Vitreo-retinal traction during inertial loading is the predominant model for severe retinal injury (Christian et al., 2018; Levin, 2010); a critical hypothesis for testing. There is no direct way to determine the biomechanics of events believed to have injured these subjects. Based on experimental studies done on primate models (Genarelli et al., 1982; Genarelli and Thibault, 1982; Ommaya and Gennarelli, 1974), we used bilateral or interhemispheric subdural hemorrhage and altered consciousness as proxies for inertial injury. The concept of inertial injury assumes traumatic cause. A few children in the PediBIRN database have neither indications of inertial injuries nor of contact injuries, opening up the possibility of non-traumatic cause. As a comparator for inertial injury, we only used children with contact injuries to clearly compare trauma to trauma.

Radiographically demonstrated diffuse HIE had significant associations with severe retinal injury in our multivariable regression

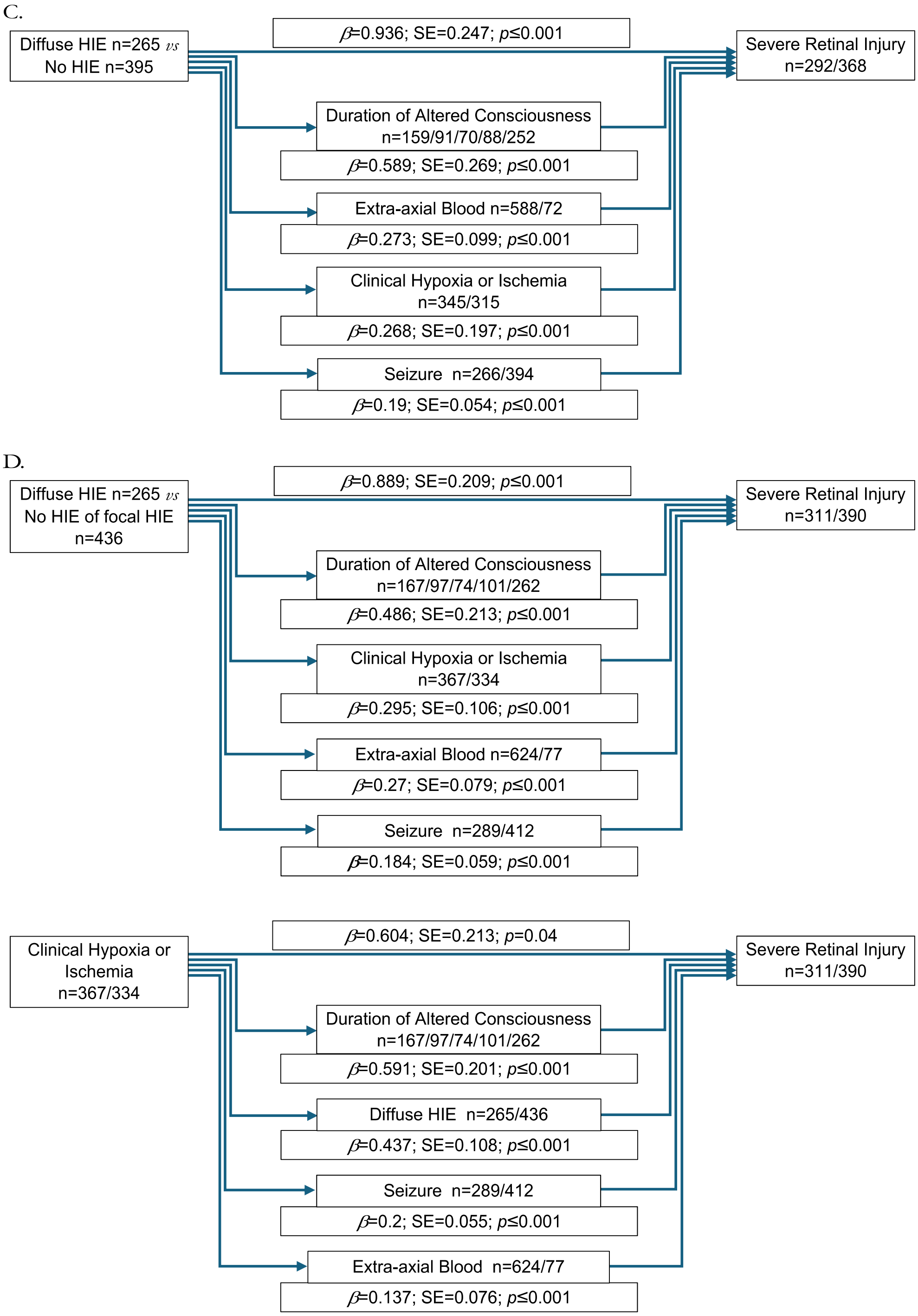

Fig. 1. (*continued*).

and was the focus of Brook's argument on causation. Increased intracranial pressure has also been raised as a potential etiology of severe retinal injury (Gabaeff, 2011; Geddes et al., 2003). The Pediatric Brain Injury Network formulation of HIE also includes brain swelling, a cause of elevated intracranial pressure. HIE, thus, serves as a proxy for severe primary brain injury and/or increased intracranial pressure. We compared diffuse HIE to cases with no HIE, and to cases where HIE was entirely absent or restricted to a focal distribution.

Hypoxemia and/or ischemia is being advanced in court as a mimic of abusive head trauma. Literature has reported a relationship between clinical hypoxia and subdural hemorrhage (Cohen and Scheimberg, 2009; Gabaeff, 2011; Geddes et al., 2003; Mack et al.,

Table 3
Results of multivariable logistic regression for the full model and comparative univariable data from Sokoloff (Sokoloff et al., 2024). $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$.

Predictor	OR (95 % CI) Sokoloff (Sokoloff et al., 2024)	OR (95 % CI) Full Model	OR (95 % CI) Select Model
Intercept	NA	0.03 (0.01–0.07) **	1.01 (0.92–1.10)
Abuse	4.51 (3.24–6.29) ***	2.06 (1.36–3.12) ***	
Age	OR not calculated $p = 0.01$	0.96 (0.94–0.99) *	0.99 (0.99–1.00) ***
Respiratory compromise	5.17 (93.73–7.15) ***	1.58 (0.94–2.65)	1.09 (0.99–1.19)
Circulatory compromise	4.30 (3.05–6.06) ***	1.80 (1.10–2.97) *	1.12 (1.03–1.22) **
Seizure	3.54 (2.58–4.86) ***	1.67 (1.10–2.54) *	1.12 (1.04–1.20) **
Altered consciousness	5.62 (3.94–8.03) ***	1.65 (0.96–2.84)	1.10 (1.00–1.20) *
Contact subdural	0.28 (0.018–0.045) ***	3.77 (1.57–9.77) **	1.14 (1.02–1.27) *
Inertial subdural	8.12 (5.44–12.12) ***	8.70 (4.13–20.34) ***	1.39 (1.27–1.53) ***
Subarachnoid hemorrhage	1.49 (1.09–2.04) **	0.94 (0.60–1.46)	
Cortical brain injury	0.90 (0.57–1.44)	0.89 (0.49–1.63)	
Subcortical brain injury	1.80 (1.01–3.23) *	0.82 (0.39–1.73)	
Focal HIE	1.09 (0.58–2.05)	1.55 (0.70–3.44)	
Diffuse HIE	5.69 (4.07–7.95) ***	2.51 (1.58–4.00) ***	1.20 (1.11–1.30) ***

2009; Scheimberg et al., 2013; Squier et al., 2012). Two papers suggested links between retinal hemorrhage and clinical hypoxia (Cohen and Scheimberg, 2009; Geddes et al., 2003). Without blood gas or oximetry data, we used reported respiratory and/or circulatory support prior to arrival in the emergency department as a proxy for clinical hypoxia or ischemia. The radiographic finding “hypoxic ischemic encephalopathy” was not treated as a proxy for poor brain oxygenation or perfusion. The etiologies of this finding are numerous, including hypoxia, ischemia, toxins, metabolic derangements and trauma (Meyers, 2021; Sharma et al., 2023).

The variables inertial injury and isolated inertial injury were studied as primary causes but not mediators. In the setting of trauma they are antecedent to any other variable. The variables diffuse HIE and clinical hypoxia or ischemia were considered both as primary and mediating variables. They have been proposed as surrogates for severity of overall neurologic injury (Brook, 2024), and as primary non-traumatic causes of severe retinal injury (Cohen and Scheimberg, 2009; Geddes et al., 2003). The mediating variable duration of altered consciousness was introduced as a proxy for the severity of brain injury of any cause.

4.3. Implications of multiple mediational analyses

While it is possible that severe retinal injury is a common outcome of multiple different causes, we proposed our five models as mutually exclusive alternatives. Each model demonstrated statistically significant effects. Our results do not exclude any of the primary etiologies considered in the multiple mediational analyses. Results for isolated inertial injury, however, appear the most robust. Isolated inertial injury had the greatest total and direct effect sizes. Mediational effects were only detected for diffuse HIE and extra-axial blood; comprising 35 % of the full effect. While diffuse HIE can be seen as mediating the causational chain, it can also be an indication of more severe inertial trauma, demonstrating that a threshold of inertial injury was crossed.

The consideration of extra-axial blood is more complicated. There is overlap between our proxy for inertial injury and extra-axial blood. Accelerative thresholds for subdural hemorrhage are higher than for accelerative altered consciousness (Thibault and Gennarelli, 1985). Thus, as with diffuse HIE, extra-axial blood may be a marker that a threshold was crossed rather than a step in the causational chain. Similarly, there are directional effects; coronal acceleration is more likely to cause altered consciousness while sagittal acceleration is more likely to cause subdural hemorrhage (Genarelli et al., 1982; Kleiven, 2003). A directional influence of inertial injury on severe retinal injury may be reflected in the mediational effect of extra-axial blood.

Including patients with contact injuries in addition to inertial injury decreased total and the direct effects. Mediators were extra-axial blood, clinical hypoxia or ischemia, diffuse HIE and seizure. Considerations are similar to those for isolated inertial injury.

Total and direct causational effects were further diminished for diffuse HIE. Results were similar for both comparators. >50 % of effects were mediated by duration of altered consciousness, extra-axial blood, clinical hypoxia or ischemia, and seizure. Other than extra-axial blood, each of these is a possible manifestation of traumatic brain injury. Unlike traumatic cause, diffuse HIE is intrinsically secondary, inviting consideration of antecedents to HIE.

For some authors, the explanatory insult for severe retinal injury is clinical hypoxia or ischemia (Cohen and Scheimberg, 2009; Gabaeff, 2011; Geddes et al., 2003; Mack et al., 2009; Scheimberg et al., 2013; Squier et al., 2012). Clinical hypoxia or ischemia generated the lowest total and direct effect sizes of our analyses. The mediators duration of altered consciousness, diffuse HIE, and seizure may be seen as intervening events in the causational chain or indicators of severe clinical hypoxia or ischemia. Those who believe that extra-axial blood can be initiated by clinical hypoxia or ischemia alone will see subdural hemorrhage as another consequence of severe clinical hypoxia or ischemia. Based on the quality of the supporting evidence and refuting literature (Hurley et al., 2010; Kelly et al., 2014; Rafaat et al., 2008), we disagree, seeing extra-axial blood as an indicator of physical trauma within the causational chain.

4.4. Limitations

We acknowledge the most obvious limitations of this study. The Pediatric Brain Injury Research Network captured cross-sectional data at a single point in time. Such data cannot prove causation. While we utilized multiple mediational methods that have been promoted for causational analysis in cross-sectional data (Hayes, 2017; Preacher, 2015), our results are best seen as support for an inference, rather than proof that inertial trauma is the most probable cause for severe retinal injury. True test of causation requires the use of animal models and biomechanical study – a body of literature which already exists and supports vitreoretinal traction during acceleration as the cause of severe retinal injury (Bonfiglio et al., 2015; Coats et al., 2010; Finnie et al., 2010; Rangarajan et al., 2009; Yoshida et al., 2014). Another limitation of the study is the use of proxies. We used the best proxies we could when primary measures of important concepts were not available; given that histories in cases where abuse is considered have uncertain informative value, this seemed reasonable. The most important of these is the inference that alteration of consciousness prior to ED arrival or the presence of interhemispheric or bilateral subdural hemorrhage accurately indicates inertial trauma biomechanics. Invalidation of this proxy certainly nullifies any conclusion regarding the relationship between trauma biomechanics and severe retinal injury, and likely limits our conclusion that some of these children suffered mechanical trauma at all.

The Pediatric Brain Injury Research Network benefited from 18 centers across the United States submitting prospective pre-defined data elements to a central, structured data set. This contributed to the strengths of this study: a large sample size and prospectively collected, uniform, pre-defined data. While necessary for obtaining uniform data on a large sample, the entry of data by multiple clinicians and centers cannot be tightly controlled. Differences in interpretation between clinicians, between centers, and over time may affect the data.

5. Conclusions

Variables in the Pediatric Brain Injury Research Network data set co-vary with severe retinal injury, limiting the validity of univariable analyses (Boos et al., 2022; Burkhart et al., 2015; Morad et al., 2002). We used multiple mediational analysis to test causational hypotheses for three major causational theories, while retaining the ability to consider multiple variables in potential causational chains. Based on superior predictive robustness and simplicity, we infer that primary causation by inertial biomechanics is the best current model for severe retinal injury. Each model produced statistically significant results, however; we cannot fully exclude alternate hypotheses. If accepted, the inertial model would support inferring an accelerative traumatic event of elevated severity when children manifest severe retinal injuries. Conclusions of abuse would then rest on a comparison with available clinical history and differential diagnoses that modify the physiological response to trauma.

Role of funder/sponsor

The National Institutes of Health and Pennsylvania State University had no role in the design or conduct of the study; the collection, management, analysis, or interpretation of the data; the preparation, review, or approval of the manuscript; or the decision to submit the manuscript for publication. The content of this study is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health or Pennsylvania State University.

CRediT authorship contribution statement

Stephen C. Boos: Visualization, Supervision, Project administration, Methodology, Conceptualization, Writing – original draft. **Fang Wang:** Visualization, Methodology, Formal analysis, Writing – review & editing. **Ming Wang:** Visualization, Supervision, Methodology, Formal analysis, Writing – review & editing. **Kent P. Hymel:** Methodology, Data curation, Conceptualization, Writing – review & editing.

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Declaration of competing interest

None.

Data availability

The data that has been used is confidential.

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